

Differential Equations and Slope Fields

ANSWER KEY



Verifying solutions to differential equations

- 1 Verify that $y = Ce^{3x}$ is a solution to the differential equation $\frac{dy}{dx} = 3y$ for any constant C .
 $\frac{dy}{dx} = 3Ce^{3x} = 3(Ce^{3x}) = 3y$. This confirms $y = Ce^{3x}$ satisfies the equation for any C .
- 2 Verify that $y = x^2 + C$ is a general solution to $\frac{dy}{dx} = 2x$, and find the particular solution satisfying $y(1) = 5$.
 $\frac{dy}{dx} = 2x$ ✓. Using $y(1) = 5$: $1 + C = 5 \Rightarrow C = 4$. Particular solution: $y = x^2 + 4$.

Solving separable differential equations

- 3 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
 $y dy = x dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 3$: $9 = 0 + 2C \Rightarrow 2C = 9$. Solution: $y^2 = x^2 + 9$, i.e. $y = \sqrt{x^2 + 9}$ (positive root, since $y(0) = 3 > 0$).
- 4 Solve $\frac{dy}{dx} = 2xy$ given $y(0) = 1$.
 $\frac{1}{y} dy = 2x dx \Rightarrow \ln|y| = x^2 + C$. Using $y(0) = 1$: $\ln 1 = 0 + C \Rightarrow C = 0$. So $\ln y = x^2 \Rightarrow y = e^{x^2}$.
- 5 Solve $\frac{dy}{dx} = \frac{\cos x}{y^2}$ given $y(0) = 2$.
 $y^2 dy = \cos x dx \Rightarrow \frac{y^3}{3} = \sin x + C$. Using $y(0) = 2$: $\frac{8}{3} = 0 + C \Rightarrow C = \frac{8}{3}$. So $y^3 = 3\sin x + 8$, i.e. $y = \sqrt[3]{3\sin x + 8}$.

Slope fields

- 6 For the differential equation $\frac{dy}{dx} = x - y$, find the slope at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, as would be drawn on a slope field.
At $(0, 0)$: slope = $0 - 0 = 0$. At $(1, 0)$: slope = $1 - 0 = 1$. At $(1, 1)$: slope = $1 - 1 = 0$.
- 7 Explain how a slope field for $\frac{dy}{dx} = ky$ (exponential growth/decay) would look different for $k > 0$ versus $k < 0$.

For $k > 0$: slopes are positive above the x -axis and negative below, steepening as $|y|$ increases — curves grow away from $y = 0$ (unstable equilibrium, exponential growth). For $k < 0$: slopes point toward $y = 0$ from both sides, flattening as $|y| \rightarrow 0$ — curves decay toward $y = 0$ (stable equilibrium, exponential decay).

Exponential growth and decay models

- 8 A bacteria culture grows according to $\frac{dP}{dt} = 0.2P$, with $P(0) = 500$. Find $P(t)$ and the population after 10 hours.

Solution form: $P(t) = P_0 e^{kt} = 500e^{0.2t}$. $P(10) = 500e^2 \approx 500(7.389) \approx 3694$ bacteria.

- 9 A radioactive substance decays according to $\frac{dA}{dt} = -0.05A$. If $A(0) = 100$ grams, find how long it takes to decay to 50 grams (the half-life).

$A(t) = 100e^{-0.05t}$. Set $50 = 100e^{-0.05t}$: $0.5 = e^{-0.05t}$, $\ln(0.5) = -0.05t$, $t = \frac{\ln(0.5)}{-0.05} = \frac{-0.693}{-0.05} \approx 13.86$ years.

Matching slope fields to differential equations

- 10 A slope field shows horizontal segments (slope 0) along the entire x -axis and along the entire y -axis, with slopes elsewhere resembling the product xy . Identify the differential equation this slope field represents.

$\frac{dy}{dx} = xy$ — slope is 0 whenever $x = 0$ or $y = 0$, matching the described field.

Newton's Law of Cooling

- 11 An object cools according to $\frac{dT}{dt} = -k(T - 70)$, where 70°F is room temperature. If $T(0) = 200^\circ\text{F}$ and $k = 0.1$, find $T(t)$.

Let $y = T - 70$: $\frac{dy}{dt} = -0.1y \Rightarrow y = y_0 e^{-0.1t}$. $y_0 = 200 - 70 = 130$. So $T(t) = 70 + 130e^{-0.1t}$.

Visualizing an exponential growth solution

- 12 The graph shows the bacteria population $P(t) = 500e^{0.2t}$ from the earlier problem. Use the graph to estimate how long it takes for the population to double from its initial value of 500, then verify algebraically using $\ln 2$.

The graph crosses $P = 1000$ at approximately $t \approx 3.5$. Algebraically: $1000 = 500e^{0.2t}$ gives $2 = e^{0.2t}$, so $t = \frac{\ln 2}{0.2} \approx 3.47$, matching the graphical estimate.

A logistic-style differential equation

- 13 A population satisfies $\frac{dP}{dt} = 0.1P(1 - \frac{P}{200})$. Find the equilibrium solutions (where $\frac{dP}{dt} = 0$), and explain which one is stable for a population starting near $P = 10$.

Setting $\frac{dP}{dt} = 0$: $P = 0$ or $P = 200$. Starting near $P = 10$ (close to 0 but positive), the population will grow toward the stable equilibrium $P = 200$ (the carrying capacity), since $P = 0$ is unstable for any positive starting population.

Additional separable equations

- 14 Solve $\frac{dy}{dx} = \frac{y}{x}$ given $y(1) = 6$.

$\frac{1}{y}dy = \frac{1}{x}dx \Rightarrow \ln|y| = \ln|x| + C \Rightarrow y = Ax$ for a constant A . Using $y(1) = 6$: $A = 6$: $y = 6x$.

- 15 Solve $\frac{dy}{dx} = \frac{1}{y}$ given $y(0) = 4$.

$y dy = dx \Rightarrow \frac{y^2}{2} = x + C$. Using $y(0) = 4$: $8 = 0 + C \Rightarrow C = 8$. So $y^2 = 2x + 16$, i.e. $y = \sqrt{2x + 16}$ (positive root, since $y(0) = 4 > 0$).

A mixing problem modeled by a differential equation

- 16 A tank contains 50 L of pure water. Brine with concentration 3 kg/L flows in at 2 L/min, and the well-mixed solution drains at the same rate. If $y(t)$ is the amount of salt (kg) at time t , the differential equation is $\frac{dy}{dt} = 6 - \frac{2y}{50} = 6 - \frac{y}{25}$. Find the equilibrium value of y , and explain what it represents physically.

Setting $\frac{dy}{dt} = 0$: $6 = \frac{y}{25}$ gives $y = 150$ kg. This represents the long-run salt content: eventually the rate of salt entering (6 kg/min from the inflow) balances the rate leaving, keeping the amount constant at 150 kg.

Multiple choice — easy 15

- 17 For the differential equation $\frac{dy}{dx} = -3x + 5y$, find the slope at the point $(-1, -4)$, as would be drawn on a slope field.

- a. -19
- b. -17 ✓
- c. -15
- d. 23

Slope = $-3(-1) - 20 = 3 - 20 = -17$.

- 18 For the differential equation $\frac{dy}{dx} = -3x - 5y$, find the slope at the point $(-2, 2)$, as would be drawn on a slope field.

- a. -2
- b. -4 ✓
- c. -6
- d. 16

$$\text{Slope} = -3(-2) - 10 = 6 - 10 = -4.$$

- 19 For the differential equation $\frac{dy}{dx} = -5x$, find the slope at the point $(0, -3)$, as would be drawn on a slope field.

- a. -2
- b. 1
- c. 0 ✓
- d. 2

$$\text{Slope} = -5(0) = 0 = 0.$$

- 20 For the differential equation $\frac{dy}{dx} = 4x - 2y$, find the slope at the point $(-4, 4)$, as would be drawn on a slope field.

- a. -22
- b. -26
- c. -24 ✓
- d. -8

$$\text{Slope} = 4(-4) - 8 = -16 - 8 = -24.$$

- 21 For the differential equation $\frac{dy}{dx} = -x - 3y$, find the slope at the point $(3, 3)$, as would be drawn on a slope field.
- a. 6
 - b. -14
 - c. -10
 - d. -12 ✓

$$\text{Slope} = -1(3) - 9 = -3 - 9 = -12.$$

- 22 For the differential equation $\frac{dy}{dx} = 3x - 5y$, find the slope at the point $(3, -2)$, as would be drawn on a slope field.
- a. 17
 - b. 19 ✓
 - c. -1
 - d. 21

$$\text{Slope} = 3(3) + 10 = 9 + 10 = 19.$$

- 23 For the differential equation $\frac{dy}{dx} = -3x + 5y$, find the slope at the point $(3, 1)$, as would be drawn on a slope field.
- a. -4 ✓
 - b. -14
 - c. -6
 - d. -2

$$\text{Slope} = -3(3) + 5 = -9 + 5 = -4.$$

- 24 For the differential equation $\frac{dy}{dx} = 5x - y$, find the slope at the point $(-4, -3)$, as would be drawn on a slope field.

- a. -15
- b. -19
- c. -17 ✓
- d. -23

$$\text{Slope} = 5(-4) + 3 = -20 + 3 = -17.$$

- 25 For the differential equation $\frac{dy}{dx} = -5x + y$, find the slope at the point $(-2, -3)$, as would be drawn on a slope field.

- a. 13
- b. 7 ✓
- c. 9
- d. 5

$$\text{Slope} = -5(-2) - 3 = 10 - 3 = 7.$$

- 26 For the differential equation $\frac{dy}{dx} = 5x + y$, find the slope at the point $(-4, -4)$, as would be drawn on a slope field.

- a. -24 ✓
- b. -16
- c. -26
- d. -22

$$\text{Slope} = 5(-4) - 4 = -20 - 4 = -24.$$

- 27 For the differential equation $\frac{dy}{dx} = 0x + 2y$, find the slope at the point $(4, -4)$, as would be drawn on a slope field.

- a. -10
- b. 8
- c. -8 ✓
- d. -6

$$\text{Slope} = 0(4) - 8 = 0 - 8 = -8.$$

- 28 For the differential equation $\frac{dy}{dx} = -2x - 4y$, find the slope at the point $(0, -4)$, as would be drawn on a slope field.

- a. 18
- b. -16
- c. 16 ✓
- d. 14

$$\text{Slope} = -2(0) + 16 = 0 + 16 = 16.$$

- 29 For the differential equation $\frac{dy}{dx} = -3x + 2y$, find the slope at the point $(-4, 4)$, as would be drawn on a slope field.

- a. 20 ✓
- b. 18
- c. 4
- d. 22

$$\text{Slope} = -3(-4) + 8 = 12 + 8 = 20.$$

- 30 For the differential equation $\frac{dy}{dx} = 0x + 2y$, find the slope at the point $(4, 4)$, as would be drawn on a slope field.

a. 8 ✓

b. -8

c. 10

d. 6

$$\text{Slope} = 0(4) + 8 = 0 + 8 = 8.$$

- 31 For the differential equation $\frac{dy}{dx} = -x + 2y$, find the slope at the point $(-4, -2)$, as would be drawn on a slope field.

a. 8

b. 0 ✓

c. -2

d. 2

$$\text{Slope} = -1(-4) - 4 = 4 - 4 = 0.$$

Multiple choice — medium 15

- 32 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 21$. Find $y(20)$ (the positive root).

a. 21

b. 29 ✓

c. 20

d. 28

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 21: 2C = 441. \text{ So } y^2 = x^2 + 441;$$
$$y(20) = \sqrt{(20)^2 + 441} = \sqrt{841} = 29.$$

33 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(-9)$ (the positive root).

- a. 12
- b. 9
- c. 14
- d. 15 ✓

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 12: 2C = 144. \text{ So } y^2 = x^2 + 144;$$
$$y(-9) = \sqrt{(-9)^2 + 144} = \sqrt{225} = 15.$$

34 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 24$. Find $y(18)$ (the positive root).

- a. 30 ✓
- b. 18
- c. 29
- d. 24

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 24: 2C = 576. \text{ So } y^2 = x^2 + 576;$$
$$y(18) = \sqrt{(18)^2 + 576} = \sqrt{900} = 30.$$

35 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 8$. Find $y(-15)$ (the positive root).

- a. 17 ✓
- b. 16
- c. 8
- d. 15

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 8: 2C = 64. \text{ So } y^2 = x^2 + 64;$$
$$y(-15) = \sqrt{(-15)^2 + 64} = \sqrt{289} = 17.$$

36 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 40$. Find $y(-9)$ (the positive root).

a. 40

b. 39

c. 41 ✓

d. 9

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 40: 2C = 1600. \text{ So } y^2 = x^2 + 1600;$$
$$y(-9) = \sqrt{(-9)^2 + 1600} = \sqrt{1681} = 41.$$

37 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 7$. Find $y(24)$ (the positive root).

a. 7

b. 25 ✓

c. 27

d. 24

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 7: 2C = 49. \text{ So } y^2 = x^2 + 49;$$
$$y(24) = \sqrt{(24)^2 + 49} = \sqrt{625} = 25.$$

38 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(9)$ (the positive root).

a. 9

b. 12

c. 15 ✓

d. 14

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 12: 2C = 144. \text{ So } y^2 = x^2 + 144;$$
$$y(9) = \sqrt{(9)^2 + 144} = \sqrt{225} = 15.$$

39 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 24$. Find $y(10)$ (the positive root).

a. 10

b. 25

c. 26 ✓

d. 24

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 24: 2C = 576. \text{ So } y^2 = x^2 + 576;$$
$$y(10) = \sqrt{(10)^2 + 576} = \sqrt{676} = 26.$$

40 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 7$. Find $y(-24)$ (the positive root).

a. 25 ✓

b. 28

c. 7

d. 24

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 7: 2C = 49. \text{ So } y^2 = x^2 + 49;$$
$$y(-24) = \sqrt{(-24)^2 + 49} = \sqrt{625} = 25.$$

41 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 21$. Find $y(-20)$ (the positive root).

a. 21

b. 29 ✓

c. 28

d. 20

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 21: 2C = 441. \text{ So } y^2 = x^2 + 441;$$
$$y(-20) = \sqrt{(-20)^2 + 441} = \sqrt{841} = 29.$$

42 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 20$. Find $y(15)$ (the positive root).

a. 24

b. 25 ✓

c. 20

d. 15

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 20: 2C = 400. \text{ So } y^2 = x^2 + 400;$$
$$y(15) = \sqrt{(15)^2 + 400} = \sqrt{625} = 25.$$

43 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(-5)$ (the positive root).

a. 12

b. 13 ✓

c. 16

d. 5

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 12: 2C = 144. \text{ So } y^2 = x^2 + 144;$$
$$y(-5) = \sqrt{(-5)^2 + 144} = \sqrt{169} = 13.$$

44 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 4$. Find $y(3)$ (the positive root).

a. 7

b. 5 ✓

c. 4

d. 3

$$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C. \text{ Using } y(0) = 4: 2C = 16. \text{ So } y^2 = x^2 + 16; y(3) = \sqrt{(3)^2 + 16} = \sqrt{25} = 5$$

45 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 24$. Find $y(-10)$ (the positive root).

- a. 24
- b. 10
- c. 25
- d. 26 ✓

$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 24$: $2C = 576$. So $y^2 = x^2 + 576$;
 $y(-10) = \sqrt{(-10)^2 + 576} = \sqrt{676} = 26$.

46 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 16$. Find $y(12)$ (the positive root).

- a. 12
- b. 19
- c. 16
- d. 20 ✓

$y \, dy = x \, dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 16$: $2C = 256$. So $y^2 = x^2 + 256$;
 $y(12) = \sqrt{(12)^2 + 256} = \sqrt{400} = 20$.

Multiple choice — hard 15

47 A quantity grows according to $\frac{dP}{dt} = \frac{3}{20}P$, with $P(0) = 326$. Find $P(5)$, rounded to the nearest whole number.

- a. 587
- b. 594
- c. 794
- d. 690 ✓

$P(t) = P_0 e^{kt} = 326 e^{\frac{3}{20}t}$. $P(5) = 326 e^{\frac{3}{20}(5)} \approx 690$.

- 48 A quantity grows according to $\frac{dP}{dt} = \frac{3}{20}P$, with $P(0) = 425$. Find $P(12)$, rounded to the nearest whole number.

a. 2957

b. 2571 ✓

c. 2185

d. 2213

$$P(t) = P_0 e^{kt} = 425 e^{\frac{3}{20}t}. P(12) = 425 e^{\frac{3}{20}(12)} \approx 2571.$$

- 49 A quantity grows according to $\frac{dP}{dt} = \frac{3}{20}P$, with $P(0) = 220$. Find $P(11)$, rounded to the nearest whole number.

a. 1146 ✓

b. 974

c. 1317

d. 986

$$P(t) = P_0 e^{kt} = 220 e^{\frac{3}{20}t}. P(11) = 220 e^{\frac{3}{20}(11)} \approx 1146.$$

- 50 A quantity grows according to $\frac{dP}{dt} = \frac{1}{4}P$, with $P(0) = 805$. Find $P(10)$, rounded to the nearest whole number.

a. 9807 ✓

b. 8336

c. 7638

d. 11278

$$P(t) = P_0 e^{kt} = 805 e^{\frac{1}{4}t}. P(10) = 805 e^{\frac{1}{4}(10)} \approx 9807.$$

- 51 A quantity decays according to $\frac{dP}{dt} = -\frac{1}{4}P$, with $P(0) = 473$. Find $P(7)$, rounded to the nearest whole number.

- a. 106
- b. 95
- c. 70
- d. 82 ✓

$$P(t) = P_0 e^{kt} = 473e^{-\frac{1}{4}t}. P(7) = 473e^{-\frac{1}{4}(7)} \approx 82.$$

- 52 A substance decays according to $\frac{dA}{dt} = -\frac{1}{10}A$. If $A(0) = 126$ grams, find how long it takes to decay to 84 grams, to one decimal place.

- a. 9.1
- b. 2.8
- c. 4.1 ✓
- d. 5.3

$$A(t) = 126e^{-\frac{1}{10}t}. \text{ Set } 84 = 126e^{-\frac{1}{10}t}: t = \frac{\ln(\frac{2}{3})}{-\frac{1}{10}} \approx 4.1.$$

- 53 A substance decays according to $\frac{dA}{dt} = -\frac{2}{25}A$. If $A(0) = 120$ grams, find how long it takes to decay to 40 grams, to one decimal place.

- a. 9.6
- b. 13.7 ✓
- c. 18.7
- d. 17.9

$$A(t) = 120e^{-\frac{2}{25}t}. \text{ Set } 40 = 120e^{-\frac{2}{25}t}: t = \frac{\ln(\frac{1}{3})}{-\frac{2}{25}} \approx 13.7.$$

- 54 A substance decays according to $\frac{dA}{dt} = -\frac{1}{20}A$. If $A(0) = 267$ grams, find how long it takes to decay to 89 grams, to one decimal place.

a. 22.0 ✓

b. 28.6

c. 27.0

d. 15.4

$$A(t) = 267e^{-\frac{1}{20}t}. \text{ Set } 89 = 267e^{-\frac{1}{20}t}: t = \frac{\ln(\frac{1}{3})}{-\frac{1}{20}} \approx 22.0.$$

- 55 A substance decays according to $\frac{dA}{dt} = -\frac{1}{20}A$. If $A(0) = 162$ grams, find how long it takes to decay to 54 grams, to one decimal place.

a. 27.0

b. 22.0 ✓

c. 28.6

d. 15.4

$$A(t) = 162e^{-\frac{1}{20}t}. \text{ Set } 54 = 162e^{-\frac{1}{20}t}: t = \frac{\ln(\frac{1}{3})}{-\frac{1}{20}} \approx 22.0.$$

- 56 A substance decays according to $\frac{dA}{dt} = -\frac{3}{100}A$. If $A(0) = 64$ grams, find how long it takes to decay to 32 grams, to one decimal place.

a. 30.0

b. 16.2

c. 23.1 ✓

d. 28.1

$$A(t) = 64e^{-\frac{3}{100}t}. \text{ Set } 32 = 64e^{-\frac{3}{100}t}: t = \frac{\ln(\frac{1}{2})}{-\frac{3}{100}} \approx 23.1.$$

57 A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 9 - \frac{y}{25}$. Find the equilibrium value of y .

a. 216

b. 34

c. 225 ✓

d. 250

Setting $\frac{dy}{dt} = 0$: $9 = \frac{y}{25}$ gives $y = 9(25) = 225$ kg.

58 A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 5 - \frac{y}{35}$. Find the equilibrium value of y .

a. 170

b. 40

c. 175 ✓

d. 210

Setting $\frac{dy}{dt} = 0$: $5 = \frac{y}{35}$ gives $y = 5(35) = 175$ kg.

59 A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 10 - \frac{y}{30}$. Find the equilibrium value of y .

a. 300 ✓

b. 290

c. 330

d. 40

Setting $\frac{dy}{dt} = 0$: $10 = \frac{y}{30}$ gives $y = 10(30) = 300$ kg.

60 A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 2 - \frac{y}{15}$. Find the equilibrium value of y .

a. 30 ✓

b. 17

c. 45

d. 28

Setting $\frac{dy}{dt} = 0$: $2 = \frac{y}{15}$ gives $y = 2(15) = 30$ kg.

61 A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 5 - \frac{y}{57}$. Find the equilibrium value of y .

a. 280

b. 62

c. 285 ✓

d. 342

Setting $\frac{dy}{dt} = 0$: $5 = \frac{y}{57}$ gives $y = 5(57) = 285$ kg.